

GWYNETH PHANG

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Education

Imperial College London, MSc Computing Sep 2024 – Sep 2025

- Software Systems Engineering: **Azure** app containerisation, deploying **Docker** containers, **PostgreSQL** databases
- Machine Learning: Implement decision tree classifiers, artificial NN for regression (**PyTorch**)
- Databases: Concurrency control, recovery, functional dependencies & normalisation, Datalog, ER modelling
- Graphics: ShaderLabWeb (framework built on **OpenGL**) for raytracing, texture, illumination rendering
- Vision: Image filtering & edge detection (2D Gaussian, Sobel), brain image segmentation (U-Net)

University of Michigan - Ann Arbor, B.S. Biopsychology, Cognition, and Neuroscience May 2022 – May 2024

- GPA: 3.9/4.0 (Distinction, James B. Angell Scholar, University Honors)
- Statistical Modelling & Data Visualization (R): normal distributions, Poisson, AIC
- Developmental Psychology: independent research, bivariate correlation tests on data (IBM SPSS)
- Computer Science Minor: Software Engineering, Data Structures & Algorithms

Projects

Psychotherapy App: Recognising and Modelling Emotions on Avatar gphang/ar-psychotherapy
Imperial College London, Algorithmic Human Development Group (AHD)

- Developed iOS application on **Unity (C#)** for self-administrable Self-Initiated Humour Protocol psychotherapy
- **Metaperson AvatarSDK API** to render 3D therapist & generate child avatar from user-submitted childhood photos
- Built chatbot interface using **Gemini LLM (Ollama)** for local deployment, enabling users to receive tailored exercises
- **Fine-tuned RoBERTa ML** model to classify emotions from text input (~95% accuracy & macro F1 score), deployed via **Flask**, **Docker** and **Google Cloud Run** for seamless application integration
- Integrated Google's real-time **Speech-to-Text** and **Text-to-Speech API** for natural voice-based interaction
- Real-time display of user emotion on avatars by editing facial expressions blendshape weights & Mixamo animations
- Implemented **AR** functionality to visualise the childhood avatar in real-world setting
- Deployed iOS application using **Xcode** & **TestFlight** for beta testing, **A/B testing** to evaluate emotion display

Gamified Storytelling App: Cloud Engineering and Full-Stack Development gphang/noko-web
Amazon Web Services (AWS), University Engagement Program

- Designed picture book application to promote cognitive development in children from disadvantaged communities
- Store picture book & user details in **DynamoDB**, with backend retrieval deployed on **AWS Lambda** for scalability
- Built **HTML** and **JavaScript** frontend pages deployed on **GitHub Pages** for a seamless, low-cost website solution
- **Text-to-Speech** using Web Speech API, with word highlighting and customisable reading speed for optimal learning
- Secure user authentication and account management with **AWS Cognito** to enhance data privacy and user trust
- Quizzes based on developmental psychology research, reward system linked to user's reading progress & quiz results

Experiences

Fixed Income Intern, Primas Asset Management Limited – Hong Kong Aug 2024 – Aug 2024

- Developed interface for bond universe filtration through **Bloomberg BQuant** IDE (Jupyter Notebook)
- Improved efficiency for capturing price movement, filtering & sorting from Bloomberg & **Python** instead of VBA
- Reduced time needed to generate report from hours to seconds, now allowing daily monitoring (previously weekly)
- Developed intuitive **front-end** page (ipydatagrid & ipywidgets), using **back-end pandas** dataframes & manipulation
- Designed 10Y Yield Prediction model, building upon Bloomberg's BECO model equation for improved accuracy
- Proposed training RLHF model with geopolitical & Fed-speak data for sentiment score (10Y Yield predictive factor)

Quantitative Developer, UBS LPoA Portfolio Manager – Singapore Dec 2022 – Jul 2024

- Created a quantitative model used for systematic trading of 40% equity allocation of US\$117 million portfolio
- Generated **automated predictive signals** to enable intraday algorithmic trading of equity index ETF
- **Statistical analysis** of momentum indicators to find parameters with highest correlation to price movement
- Model capable of capturing investor sentiments, timing breakout inflexion points at the start of trendline
- Programmed strategy into TradingView using **PineScript** language to generate real-time weighted buy/sell signals
- 12-month record of +35% profit -2% drawdown for QQQ, +18% profit -3% drawdown for SPY, beating benchmarks

Skills

UBS LEADS Program: Led startup pitch for MVP (business modelling, competitive analysis, investor relations)

Google User Experience Design Professional Cert: UX research, create wireframes, low & high-fidelity prototypes

Programming Languages: C++, C#, C, Python, GLSL, JavaScript, HTML, SQL, Datalog, R

Technologies: VSCode, Xcode, Unity, OpenGL, Azure, AWS, Docker, PostgreSQL, BQuant, Tableau, IBM SPSS Statistics

Foreign Languages: Mandarin, Cantonese, Japanese